

MACE, DFT and the Cu–Au demo

A presenter's primer: what the physics is, what the model is, what the demo does, and how to answer the questions

ENGR 689 tutorial · demo repository github.com/soniyuvraj16/mace-cuau-demo · numbers quoted are from the actual run on real VASP-PBE labels

1. The one idea everything hangs on
 2. DFT basics: what, why trusted, what's wrong with it
 3. From DFT to machine-learned potentials
 4. MACE specifically
 5. Fine-tuned vs raw MACE
 6. The Cu–Au system
 7. How the demo shows materials discovery
 8. Question bank
 9. The 60-second opener and the numbers to remember
- Appendix: our exact settings and references

1. The one idea everything hangs on

Almost every materials property follows from one function: the **potential energy of a set of atoms as a function of their positions and species**, $E(\mathbf{r}_1 \dots \mathbf{r}_N, Z_1 \dots Z_N)$, and its gradient, the **forces**. Which crystal structure forms, whether an alloy orders or separates, how atoms move at temperature, how an electrode swells — all of it is this function evaluated many times. The whole field is a trade-off between *how accurately* you compute it and *how many times* you can afford to.

Method	How it gets the energy	Cost per evaluation	Accuracy / transferability
Quantum mechanics (DFT)	Solves for the electrons	Minutes to hours	High; first principles; works for any element
Classical force fields (Lennard-Jones, EAM)	Hand-written formula with fitted parameters	Microseconds	Low outside what it was fitted to; often qualitatively wrong for new chemistry
Machine-learned potentials (MACE)	Neural network trained to reproduce DFT	Milliseconds	Near-DFT inside its training distribution; must be checked outside it

Framing for a CS/EE room: an expensive oracle (DFT), a cheap learned surrogate (MACE), and a loop that uses the surrogate for search and the oracle for verification. Everyone who has seen Bayesian optimisation or surrogate modelling already knows this story; the materials part is just the instance.

2. DFT basics

What it is

The exact description of electrons in a material is the many-electron Schrödinger equation: a wavefunction of $3N$ coordinates for N electrons, hopeless to solve directly beyond a handful of electrons. **Density functional theory** (Hohenberg & Kohn 1964; Kohn & Sham 1965; Kohn's 1998 Nobel Prize) proves that the ground-state energy is a *functional of the electron density* $n(\mathbf{r})$ — a function of just three coordinates — and recasts the problem as non-interacting electrons moving in an effective potential that depends on the

density itself. The equations are solved iteratively until density and potential agree: the self-consistent-field (SCF) loop. Our `EDIFF = 1e-6` is the convergence threshold of that loop.

The one approximation

Everything is exact except the **exchange–correlation functional**, the term capturing how electrons avoid each other. Its exact form is unknown; the standard approximations are LDA and GGA. We used **PBE** (Perdew, Burke, Ernzerhof 1996), the most common GGA for solids, and the functional behind the Materials Project and therefore behind the MACE foundation model. Our DFT and the pretrained network speak the same dialect — relevant later.

Why it is trusted

- **First principles.** Inputs are atomic numbers and positions; nothing is fitted to experiment for the system at hand. That is what makes it transferable to elements, structures and conditions nobody has measured.
- **It captures electronic structure** — bonding, charge transfer, magnetism, metallicity — which classical potentials cannot.
- **Typical PBE accuracy for solids:** lattice constants within 1–2 % (ours: Cu 3.65 Å, Au 4.18 Å vs experiment 3.61 / 4.08 — PBE overestimates slightly), formation energies to a few tens of meV/atom, and forces analytically via the Hellmann–Feynman theorem at no extra cost.

Its problems — the reason MACE exists

1. **Cost.** Roughly cubic in the number of electrons. Our 4–8-atom single points took minutes each on 12 cores; 100 atoms is hours; a nanosecond of molecular dynamics is a million force evaluations. Any loop needing thousands of evaluations cannot have DFT inside it.
2. **"Accurate" with known systematic errors.** PBE underestimates band gaps, mishandles van der Waals forces, and *underbinds* noble-metal alloys: our Cu₃Au formation energy of –47 meV/atom is weaker than experiment (roughly –70 meV/atom). A model trained on PBE inherits PBE's errors exactly; it can never be better than its teacher.
3. **Zero kelvin.** Plain DFT energies are static. Temperature (vibrations, disorder) needs statistical mechanics on top — again, many evaluations.

Terms from our setup

Term	Meaning	Ours
VASP	The DFT code (plane-wave, periodic)	6.3.2 on TAMU HPRC Grace
PAW / POTCAR	Core electrons frozen; only valence electrons treated	PBE Cu, Au, 11 valence e [–] each
ENCUT	Plane-wave basis cut-off: resolution of the wavefunctions	400 eV
k-points	Sampling of the periodic crystal's Brillouin zone; metals need dense meshes	Γ-centred 8×8×8 (4-atom cells), 4×8×8 (8-atom)
ISMEAR / SIGMA	Smearing of the Fermi surface so metals converge	Methfessel–Paxton, 0.1 eV
Single point (NSW = 0)	Energy and forces at a fixed geometry, no relaxation	All 186 calculations

Why fixed geometries: the demo compares MACE and DFT on *identical* structures, so nothing about relaxation paths can confuse the comparison. A research study would relax each candidate.

3. From DFT to machine-learned potentials

Treat $E(\text{positions, species})$ as a regression target: run DFT on a few hundred to a few thousand representative structures, then train a model to reproduce their energies and forces. Three design principles make it work:

- **Locality.** Total energy = sum of atomic energies, each depending only on neighbours within a cut-off (MACE: 6 Å). This makes the model transferable across cell sizes — ours, trained on 4- and 8-atom cells, works on any size — and linear in cost with the number of atoms.
- **Symmetry.** The energy must not change under translation, rotation, or relabelling of identical atoms; forces must rotate with the structure. This is built into the architecture, not learned from data.
- **Forces from the gradient.** Forces are the exact derivative of the model's energy (automatic differentiation), so the force field is conservative by construction — essential for stable dynamics. Each DFT structure also yields $3N$ force labels plus one energy; forces carry most of the training signal.

Lineage (one line each)

- **Behler–Parrinello (2007):** atomic-energy decomposition with hand-crafted symmetry functions feeding small neural networks.
- **GAP (Bartók et al. 2010):** the same idea with Gaussian-process regression.
- **Atomic Cluster Expansion, ACE (Drautz 2019):** a complete, systematic many-body basis for atomic environments.
- **Equivariant graph networks (NequIP, Batzner et al. 2022):** learn the features instead of hand-crafting them, with exact rotational equivariance.
- **MACE (Batatia, Kovács, Simm, Ortner, Csányi — NeurIPS 2022):** ACE's many-body products inside an equivariant message-passing network.

4. MACE specifically

The mental model

Atoms are nodes in a graph; edges connect atoms within the cut-off. In each layer every atom collects "messages" from its neighbours describing their relative positions and species, updates its own feature vector, and after the last layer a readout maps each atom's features to an atomic energy. Sum the atomic energies for the total; differentiate for forces.

What makes MACE different

- **Higher-order messages.** Older message-passing networks combine neighbours one pair at a time and need many layers to build up many-body information. MACE forms *products* of neighbour features — the ACE construction — so one message already encodes three- and four-body geometry (angles, dihedrals). Consequence: only **two layers** are needed, which is why it is both accurate and fast.
- **E(3)-equivariance.** Features are built from spherical harmonics and carry rotational type labels (irreducible representations), so rotating the input rotates the internal features and the output forces exactly — no augmentation, no approximation.
- **Receptive field.** Two layers $\times$ 6 Å cut-off: each atom "sees" 12 Å. Enough for metals and most covalent systems; not enough for long-range electrostatics (see limitations).

Foundation models

MACE-MP-0 (Batatia et al. 2023) is a MACE trained on about 1.5 million structures from Materials Project relaxation trajectories — 89 elements, PBE. It is a general-purpose potential: load it and run almost any inorganic material zero-shot. We used the *small* variant (128 channels, scalar features), the fastest. Zero-shot accuracy is impressive for a universal model and modest for any specific task — the gap fine-tuning closes.

~40 ms

per 8-atom energy on a
laptop CPU

minutes

per DFT single point on
12 cores

2 layers

6 Å cut-off, 12 Å
receptive field

89 elements

in the MACE-MP-0
training set

Training, as we did it

Start from MACE-MP-0's weights and architecture. Fit per-element reference energies (E_0 s) to our data by least squares so the absolute energy scale of our DFT matches. Minimise $100 \cdot (\text{energy error per atom})^2 + 10 \cdot (\text{force error})^2$ with Adam (learning rate 0.01, exponential moving average of weights) for 20 epochs over 135 structures (122 train / 13 validation). The 100:10 weighting is deliberate: MACE's default is 1:100, which fits forces beautifully but leaves energies — which the convex hull depends on — under-trained in a short run. It was the one hyper-parameter that mattered.

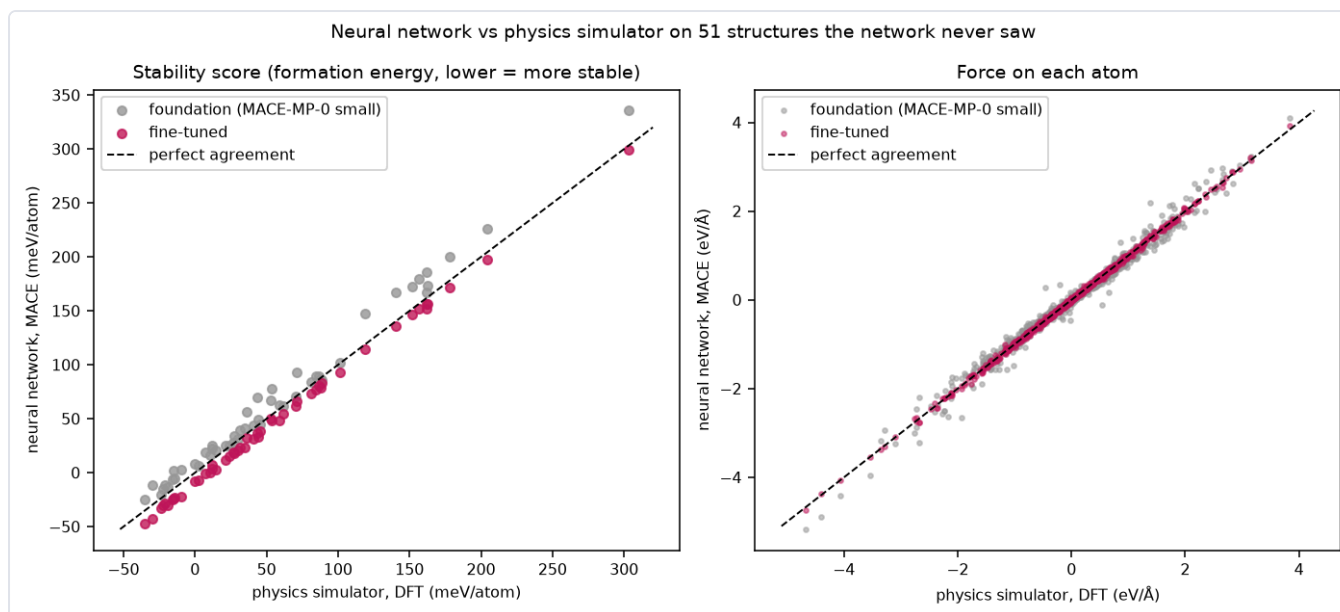
Limitations to volunteer before someone else does

- **Extrapolation.** An MLIP interpolates within the environments it has seen. Show it an arrangement unlike anything in training and it can be confidently wrong — on stand-in data we saw a never-trained 50:50 ordering over-stabilised by 9 meV/atom. This is the reason the demo has a verify step.
- **No native uncertainty.** A single model cannot tell you when it is extrapolating. The standard remedy is a committee (several models; use their disagreement), which is how active learning decides what to send to DFT next.
- **Locality.** Long-range electrostatics and dispersion are not captured by a 12 Å receptive field unless added explicitly.
- **Inherits the teacher.** PBE's systematic errors are MACE's systematic errors.

5. Fine-tuned vs raw MACE — what if we skipped step 1?

Held-out results on 51 structures of 17 orderings the model never saw:

	raw MACE-MP-0	fine-tuned, 20 epochs	fine-tuned, 30 epochs
formation-energy error (MAE)	10.7 meV/atom	7.9	~4
force error (RMSE)	112 meV/Å	26	~23



Network vs simulator on the 51 held-out structures. Grey: pretrained; pink: fine-tuned. The fine-tuned energies sit on a line parallel to perfect agreement — a uniform offset of ~ 9 meV/atom that cancels in rankings.

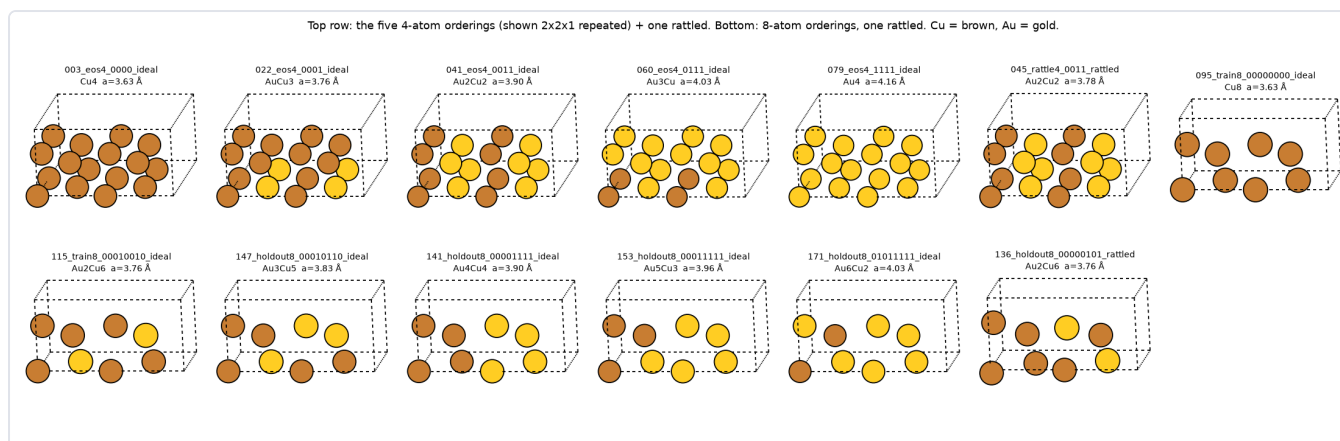
The honest reading. Raw MACE is already decent on *energies* here, because it was trained on PBE and Cu and Au are common, well-sampled elements. On *forces* it is $4\times$ worse, which is what would hurt in relaxations or dynamics. So "why not just use raw MACE?" is a fair question, with a three-part answer:

1. **The decisions live in a narrow margin.** The hull is ~ 45 meV/atom deep and rival orderings at a composition differ by single meV/atom. A 10 meV/atom error makes the raw model's ranking untrustworthy; 4 meV/atom is borderline; the fine-tuned model's systematic offset cancels in rankings, which is why its hull matches DFT's at all five vertices.
2. **Fine-tuning is how the loop learns.** Every DFT result the verify step returns can be added and the model retrained. A raw foundation model is static; a fine-tuned one improves with every expensive call. That is the "refine" step.
3. **This is the easy case.** Cu–Au at ambient pressure in small cells is squarely inside MACE-MP-0's training distribution. For new chemistries, surfaces, defects, high pressure, or a different functional or code, the raw model can be off by hundreds of meV/atom; a few hundred DFT points of fine-tuning is the difference between useless and useful.

Cost of fine-tuning: 186 DFT single points (a day on a cluster, mostly waiting) plus 3 minutes of training on a laptop.

6. The Cu–Au system — real, and historically important

Copper and gold are both face-centred cubic, mix in all proportions, and below their ordering temperatures form **ordered intermetallic compounds**: **Cu₃Au** (L_{12} structure — Au on the cube corners, Cu on the face centres; orders below ~ 390 °C), **CuAu** (L_{10} — alternating layers of Cu and Au; ~ 410 °C) and **CuAu₃** (L_{12} , more weakly ordered). The Cu₃Au order–disorder transition is the classic example in every physical-metallurgy course, and Cu–Au was the original testbed for computational ground-state searches (cluster expansion methods, 1980s–90s). Our 8-site search rediscovers L_{12} Cu₃Au and L_{10} CuAu — exactly the compounds nature forms — which is why the audience can check the answer against reality.



Top: the five 4-atom orderings used for training (shown repeated $2 \times 2 \times 1$), Cu_3Au $L1_2$ and CuAu $L1_0$ among them, plus a rattled copy. Bottom: 8-atom ($2 \times 1 \times 1$) orderings. Brown = Cu, gold = Au.

Formation energy

$E_f = E(\text{mixture}) - xE(\text{Au}) - (1-x)E(\text{Cu})$, per atom: the energy released by mixing compared with keeping the pure metals apart. Negative means the compound is favourable. Ours (PBE): Cu_3Au -46.7 , CuAu -48.6 , CuAu_3 -28.2 meV/atom. Experiment is more negative — PBE's underbinding.

Convex hull

Plot E_f against composition and draw the lowest convex envelope. Points on the hull are thermodynamically stable compounds; points above it would decompose into a mixture of their hull neighbours. "Finding the hull" is the formal definition of finding which materials exist at 0 K — it is what a ground-state search is for.

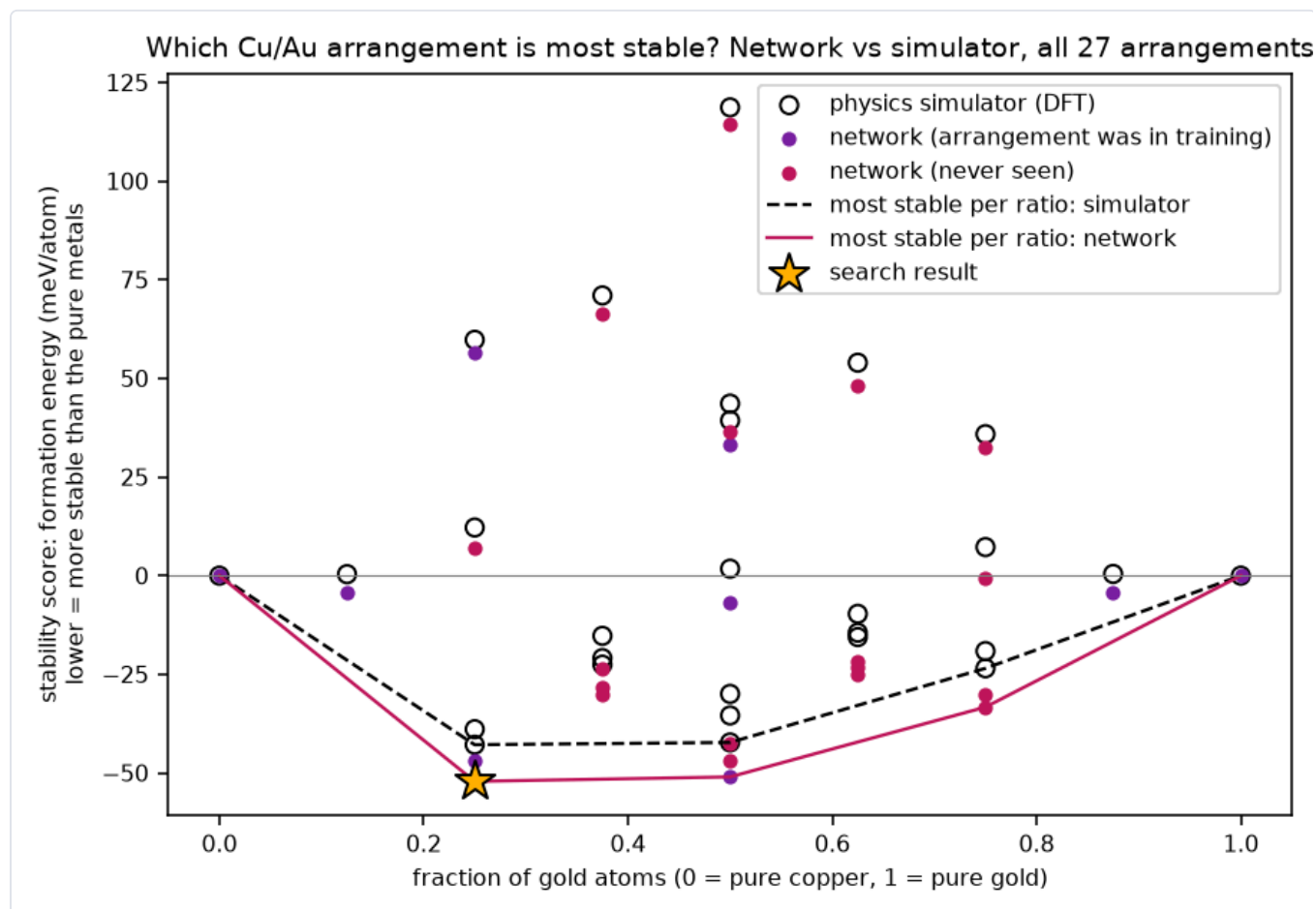
The search space, honestly

8 lattice sites, $2^8 = 256$ decorations, 27 distinct after symmetry. Small enough to enumerate — which the script also does, to show the genetic algorithm reached the true minimum with 21 evaluations. The demo shows the *mechanism*, not the scale.

A detail worth knowing if asked about the 27: an earlier version counted 25, because a "fingerprint" based on neighbour-distance histograms merged two pairs of genuinely different orderings that happen to have identical pair statistics. The fix was a proper canonical form under the FCC space-group operations. Good illustration that "distinct structures" is a symmetry question, not a geometry one.

7. How the demo shows materials discovery

Loop step	In the demo	At research scale
learn	fine-tune MACE on 135 DFT labels (3 min)	same, iterated as data arrives
propose	genetic algorithm mutates and crosses 8-bit strings	GA / random / generative models over compositions, supercells, defect sites — 10^4 – 10^6 candidates
evaluate	MACE energy, ~40 ms each	MACE relaxation (hundreds of force calls per candidate) — still seconds
decide	lowest formation energy; the hull vertices	hull distance plus target properties (gap, stability under conditions)
verify	DFT on the 3 predicted ground states instead of 27	DFT on the shortlist, triggered by committee uncertainty
refine	(would) add verified structures and retrain	active learning: retrain, search again



Step 3 output: every distinct ordering scored by the network (pink) and by DFT (open circles). The two hulls share all five vertices; the star is the GA's champion, $\text{Cu}_3\text{Au L1}_2$, confirmed by DFT as the ground state.

3 of 27

expensive calls, all three confirmed

5 / 5

hull vertices agree with DFT

21

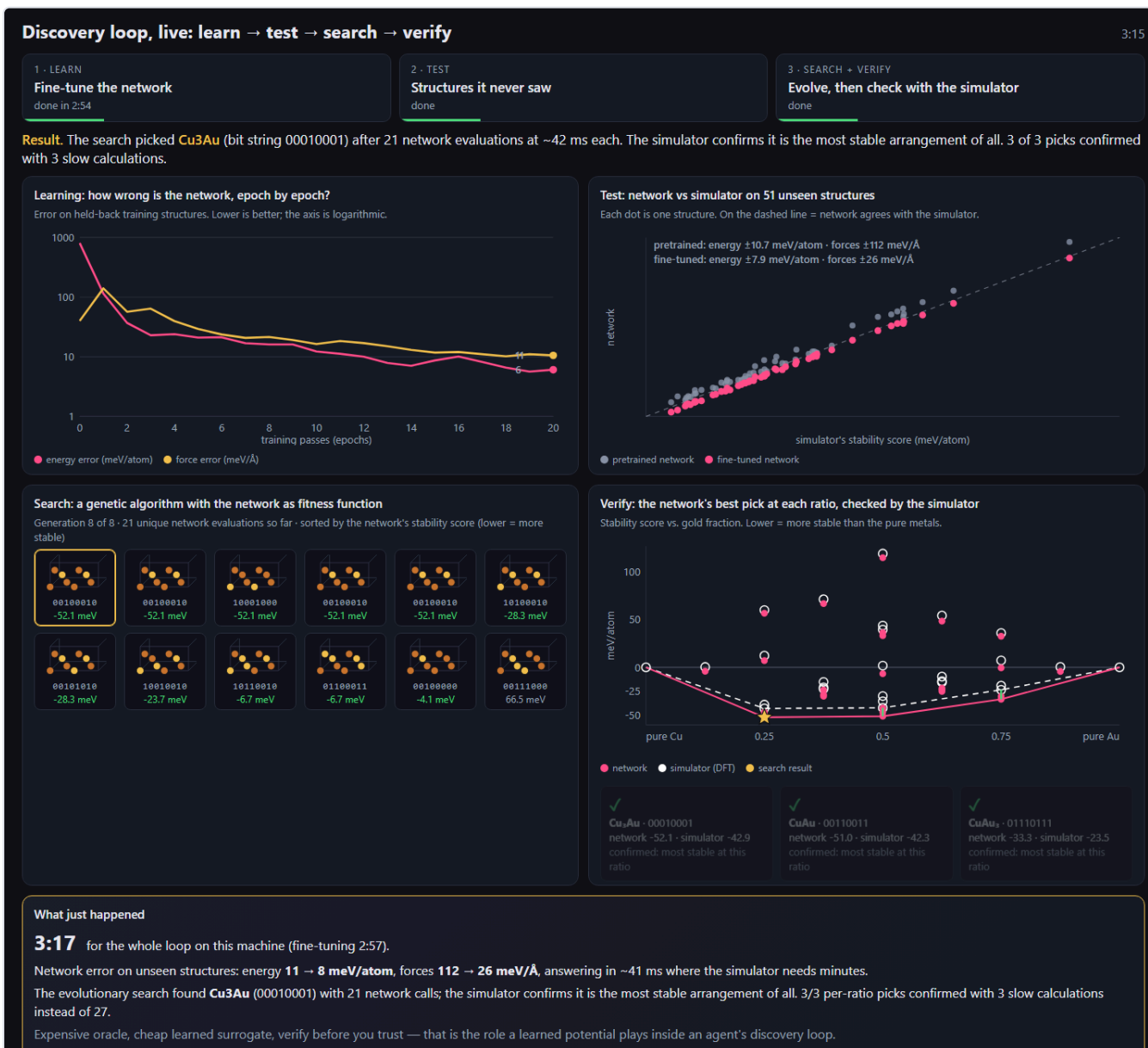
network evaluations for the GA to converge

~3.5 min

the whole loop on a laptop CPU

What the demo does not do — say it before they ask

- The champion ordering (00010001) was one of the 10 orderings in the training set. The evidence that MACE generalises is the 17 *held-out* orderings scored to 7.9 meV/atom and the hull matching at all five vertices — point there, not at the champion.
- Geometries are not relaxed and there is no temperature; a real study relaxes each candidate and runs Monte Carlo on the surrogate for order-disorder temperatures.
- MACE's energies sit ~9 meV/atom below DFT at 20 epochs. The ranking is right; the absolute numbers are not. That is exactly why the loop verifies.



The live dashboard at the end of a run: learning curve, parity cloud, the evolved population drawn as copper/gold lattices, the hull with DFT's checks, and the plain-language summary.

8. Question bank

Why not just run DFT on all 27?

For 27 you could — that is how we built the answer key. The demo is sized to run live. A 16-site cell has ~2,000 distinct orderings; add relaxation and you are at 10^5 DFT calls. The surrogate's value grows with the search space; the mechanism is identical.

How do you know when to trust the model?

Three ways, all in the demo: held-out validation on structures it never saw, the error pattern in the parity plot (systematic offset vs random scatter), and the verify step. In research you add a committee of models and trust predictions only where they agree.

Isn't a genetic algorithm overkill for 256 strings?

Yes — brute force takes 0.7 s, and the script does it to check. The GA is there because at scale enumeration is impossible; some proposal strategy (GA, Bayesian optimisation, a generative model) is what makes search scale, and the fitness function costing 40 ms instead of 10 minutes is what makes it affordable.

What is a formation energy, and what is a meV per atom?

The energy released per atom when the compound forms from the pure elements, in thousandths of an electron-volt. Room-temperature thermal energy is ~25 meV; these compounds sit 25–50 meV/atom below the elements and rivals differ by a few meV. That is the precision the model must reach.

Could MACE replace DFT?

No. It is trained on DFT and cannot exceed it. It replaces DFT *inside loops*, where thousands of evaluations are needed, and hands back to DFT for verification and for new training data.

Why PBE and not a better functional?

PBE is the standard for solids and what MACE-MP-0 was trained on, so fine-tuning starts close. Better functionals (SCAN, r^2 SCAN, hybrids) cost more per point; the workflow is unchanged — only the labels change.

What about temperature, entropy, disorder?

Everything here is 0 K. Order–disorder transitions need configurational entropy — Monte Carlo over orderings — itself a loop of millions of energy evaluations, i.e. another thing only a surrogate makes possible.

Does MACE do dynamics?

Yes; its forces are the exact gradient of its energy, so it runs stable, energy-conserving MD. The 4× force improvement from fine-tuning is what you would need for that.

What does an agent add?

The decisions: what to propose next, when to trust, when to call DFT, when to retrain. MACE is the tool; the loop is the agent's. The demo runs a fixed script of those decisions; an LLM agent would make them adaptively — for instance noticing a never-seen environment type and requesting DFT for it, which the script already flags.

How long did the DFT take?

186 single points on Grace, minutes each on 12 cores, as a job array — about a day end to end including setup. Fine-tuning on the result: 3 minutes.

Why Cu–Au?

Small cells, two well-sampled elements, a textbook answer to check against, and a hull deep enough (–45 meV/atom) that a few-meV model can resolve it.

Is the model's 9 meV/atom offset a problem?

Not for ranking — it is the same shift for every mixed composition, so the order of candidates is preserved and the hull vertices match. It would matter for absolute formation enthalpies, and 30 epochs halves it. Energies are compared as *formation* energies precisely so that per-element constant offsets between codes cancel.

What was in the 186 structures?

4-atom cells: the 5 distinct orderings, each with a 7-point volume scan (energy vs lattice constant) and 12 randomly displaced ("rattled") copies to give non-zero forces. 8-atom cells: pure Cu₈, Au₈ and 8 random mixed orderings in training (ideal + 3 rattled each); the other 17 orderings held out (ideal + 2 rattled each). An active-learning snapshot: DFT on under 40 % of the ordering space, the model ranks the rest.

9. The 60-second opener and the numbers to remember

"Everything in materials follows from one function — the energy of atoms as a function of where they are. Quantum mechanics gives it to us accurately but slowly: minutes per evaluation. Discovery needs thousands of evaluations. MACE is a neural network trained to imitate that quantum calculation at a thousandth of the cost. Today we fine-tune it on 135 quantum calculations for copper–gold, check it on 51 it never saw, let an evolutionary search use it to find the most stable arrangement, and then spend just 3 slow calculations confirming the answer — which turns out to be Cu_3Au , a compound that really exists. Expensive oracle, cheap learned surrogate, verify before you trust."

Number	What it is
186	DFT single points (135 train, 51 held out), VASP PBE
27	symmetry-distinct 8-site orderings; 10 in training, 17 never seen
10.7 → 7.9 meV/atom	held-out energy error, pretrained → fine-tuned (20 epochs); ~4 at 30 epochs
112 → 26 meV/Å	held-out force error, pretrained → fine-tuned
−46.7 / −48.6 / −28.2	PBE formation energies of Cu_3Au / CuAu / CuAu_3 , meV/atom
21	network evaluations for the GA to find Cu_3Au L1 ₂
3 instead of 27	DFT calls in the verify step; 3/3 confirmed; 5/5 hull vertices agree
~40 ms vs minutes	network vs DFT per 8-atom structure
~3.5 min	the whole loop on a laptop CPU (≈6.5 min including a fresh install; ~80 s on a small GPU)

Appendix: exact settings and references

DFT

VASP 6.3.2, PBE, PAW potentials [Cu](#) (22Jun2005) and [Au](#) (04Oct2007), ENCUT 400 eV, PREC Accurate, EDIFF 1e-6, NELM 120, ALGO Normal, ISMEAR 1 / SIGMA 0.10, ISPIN 1, LREAL .FALSE., ADDGRID, IBRION −1 / NSW 0, Γ -centred k-meshes $8\times8\times8$ (4-atom cells) and $4\times8\times8$ (8-atom cells), KPAR 4 × NCORE 3 on 12 MPI ranks per calculation, SLURM array on TAMU HPRC Grace. Lattice constants for the structures: PBE values 3.63 Å (Cu) and 4.16 Å (Au), interpolated linearly with composition (Vegard's law); volume scans $\pm 5\%$; rattles with $\sigma = 0.05\text{--}0.10$ Å.

MACE fine-tuning

Foundation model MACE-MP-0 small; single-head fine-tuning (no replay of the pre-training set); E_0 s fitted by least squares; loss weights energy 100 : forces 10; Adam with AMSGrad, learning rate 0.01, EMA 0.99; batch 16; float32; 20 epochs (default) or 30; 10 % validation split; seed 1, so CPU and GPU give the same model.

References

- Hohenberg & Kohn, *Phys. Rev.* 136, B864 (1964); Kohn & Sham, *Phys. Rev.* 140, A1133 (1965).
- Perdew, Burke & Ernzerhof, *Phys. Rev. Lett.* 77, 3865 (1996) — PBE.
- Behler & Parrinello, *Phys. Rev. Lett.* 98, 146401 (2007). Bartók et al., *Phys. Rev. Lett.* 104, 136403 (2010) — GAP. Drautz, *Phys. Rev. B* 99, 014104 (2019) — ACE. Batzner et al., *Nat. Commun.* 13, 2453 (2022) — NequIP.

- Batatia, Kovács, Simm, Ortner & Csányi, "MACE: Higher Order Equivariant Message Passing Neural Networks for Fast and Accurate Force Fields", *NeurIPS* 2022.
- Batatia et al., "A foundation model for atomistic materials chemistry", arXiv:2401.00096 (2023) — MACE-MP-0.
- Jain et al., *APL Materials* 1, 011002 (2013) — the Materials Project.
- Cu–Au ordering and first-principles alloy theory: e.g. Ozoliņš, Wolverton & Zunger, *Phys. Rev. B* 57, 6427 (1998).